Mid Semester Examination

Classical Mechanics
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Duration: 120 minutes.
Total points: 50.

Please give arguments where necessary. If it is unclear from your answer why a particular step is being taken, full credit will not be awarded. Grades will be awarded not only based on what final answer you get, but also on the intermediate steps.

1. In spherical coordinates, (r, θ, ϕ), the following Lagrangian is given by

$$L = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2 + r^2 sin^2\theta\dot{\phi}^2) - V(r, \theta)$$

   (a) What are the conserved quantities in the system in terms of the coordinates and their time-derivatives?

   (b) Specify the symmetry of the Lagrangian that led to each of the conserved quantities you mentioned above.

   (c) Physically interpret any conserved quantity that is not the total energy.

   4 + 4 + 2 = 10 points

2. Imagine a massless spring of spring constant K (which implies that its potential energy will be $V = \frac{1}{2}Kx^2$, where x is the extension of the spring beyond the equilibrium length l) is hanging (in a straight line) from a fixed horizontal platform. A mass m is attached to the bottom of the spring. The whole system can move in a vertical plane. See Figure 1.

   (a) Using the $x(t)$ and the $\theta(t)$ variables as shown in the figure as your generalized coordinates, write down the Euler-Lagrange equations of motion.

   (b) Identify the quantity that signifies the centripetal acceleration in this case. If $r(t)$ does not change with time, what does the $F = ma = 0$ condition in the radial direction look like?

   6 +(2 + 2) = 10 points

3. Consider a set of generalized coordinates (and the corresponding generalized velocities) given by $(q_1, q_2, \cdots, q_n)$ that define a Lagrangian L. We

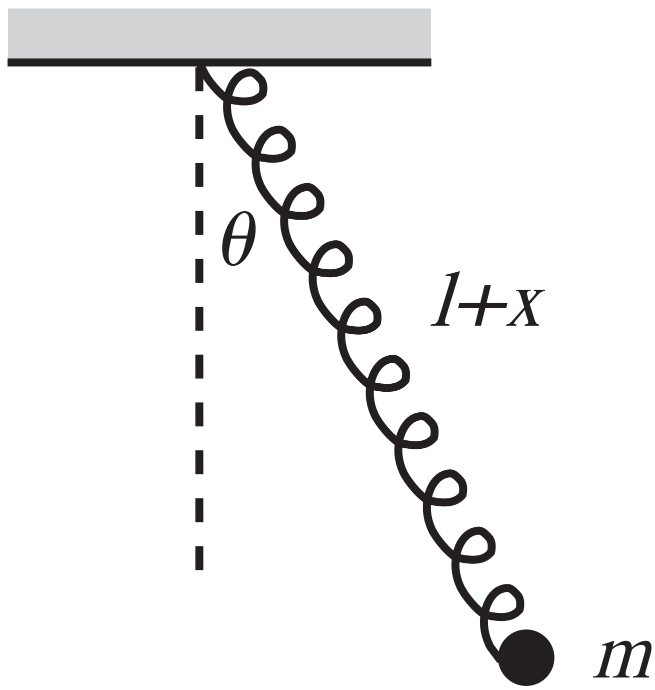

Figure 1: Mass-spring system for Problem 2

know that the actual motion of the system will be determined by the n differential Euler-Lagrange equations given by ($\forall i = 1 \cdots n$)

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\right) - \frac{\partial L}{\partial q_i} = 0$$

Imagine now that you define a new set of generalized coordinates $s_i = s_i\left(q_1, q_2, q_3, ..., q_n; t\right)$. Note that the s_i variables do not depend on the generalized velocities $\dot{q}_k$. Show that the variables s_i follow the same set of Euler-Lagrange equations $\forall i = 1 \ldots n$:

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{s}_i}\right) - \frac{\partial L}{\partial s_i} = 0$$

10 points

4. Imagine there is a wedge-shaped block of mass M which rests on a frictionless horizontal plane. The inclined plane (also frictionless) of the block is at a fixed angle θ to the horizontal. A smaller block of mass m is placed

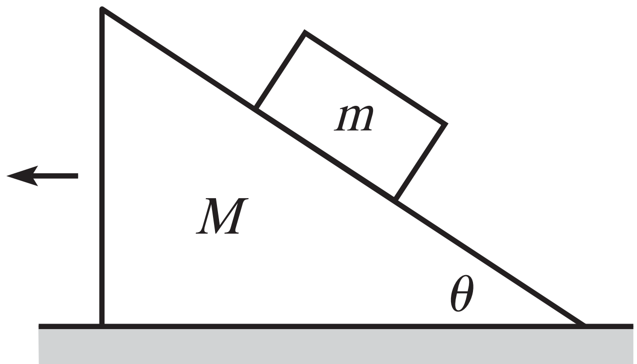

Figure 2: Block-on-block system for Problem 2

on the inclined plane of the earlier block and the system is released. See Figure 2.

(a) Find the horizontal acceleration of the M-block. The only constraint is that the smaller block stays on the surface of the larger block the entire duration.

(b) Find the angle θ at which the acceleration above is maximized, given fixed M and m.

(c) Explain physically the limits $M \ll m$ and $m \ll M$.

5 + 2 + 3 = 10 points

5. A ball of mass m is thrown as a projectile from ground level with initial speed v at an angle θ to the horizontal. It moves only under gravity, g. Find the equation for the angle θ that maximizes the trajectory length that the ball travels (Note: the trajectory length is not the horizontal distance travelled before the ball touches the ground; the trajectory length is the length measured along the actual path. The horizontal distance for projectile motion is always maximized by $\theta = \frac{\pi}{4}$).
10 points